THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 15	$\frac{10+20}{2}$; Class mark $= 15$
2	**(a)** 30–40	It belongs to the class 30–40; Modal class $=$ 30–40
3	**(b)** 40–50	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class $=$ 40–50
4	**(d)** 50	$3+9+4+18+16$; $\sum f_i = 50$
5	**(c)** 12	$3+9$; $cf = 12$
6	**(c)** 34.8	$\sum f_i = 50$ and $\sum f_i x_i = 1740$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$
7	**(a)** 58.75	$l = 50$, $f_1 = 18$, $f_0 = 4$, $f_2 = 16$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{14}{16} \times 10 = 58.75$
8	**(b)** 21	Median class 20–30: $l = 20$, $cf = 12$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{12.5-12}{5} \times 10 = 21$
9	**(b)** 30	Class limits
10	**(d)** 25	$\frac{20+30}{2}$; Class mark $= 25$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Class marks x_i: 15, 25, 35, 45; $\sum f_i = 50$ and $\sum f_i x_i = 1740$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$ (1).	1+1
12	Highest frequency is 20, so the modal class is 50–60; $l = 50$, $f_1 = 20$, $f_0 = 10$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{10}{25} \times 10 = 54$ (1).	1+1
13	Cumulative frequencies: 4, 14, 18, 25; $N = 25$, so $\frac{N}{2} = 12.5$; Median class 50–60: $l = 50$, $cf = 4$, $f = 10$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{12.5-4}{10} \times 10 = 58.5$ (1).	1+1
14	Class marks x_i: 5, 15, 25, 35; $\sum f_i = 25$ and $\sum f_i x_i = 415$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{415}{25} = 16.6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Class marks x_i: 5, 15, 25, 35, 45 (1); $\sum f_i = 50$ and $\sum f_i x_i = 970$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{970}{50} = 19.4$ (1).	1+1+1
16	Cumulative frequencies: 5, 21, 26, 39, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 40–50: $l = 40$, $cf = 21$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{25-21}{5} \times 10 = 48$ (1).	1+1+1
17	Highest frequency is 10, so the modal class is 20–30 (1); $l = 20$, $f_1 = 10$, $f_0 = 2$, $f_2 = 2$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{8}{16} \times 10 = 25$ (1).	1+1+1
	OR: Class marks x_i: 5, 15, 25, 35, 45 (1); $\sum f_i = 50$ and $\sum f_i x_i = 1240$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1240}{50} = 24.8$ (1).	1+1+1
18	Cumulative frequencies: 5, 22, 27, 45, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 70–80: $l = 70$, $cf = 22$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{25-22}{5} \times 10 = 76$ (1).	1+1+1
	OR: Highest frequency is 15, so the modal class is 20–30 (1); $l = 20$, $f_1 = 15$, $f_0 = 11$, $f_2 = 9$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{4}{10} \times 10 = 24$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (2); $\sum f_i = 50$ and $\sum f_i d_i = -510$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{-510}{50} = 74.8$ (1).	2+2+1
	OR: Cumulative frequencies: 10, 12, 18, 34, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 40–50: $l = 40$, $cf = 18$, $f = 16$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{20-18}{16} \times 10 = 41.25$ (1).	2+2+1
20	Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (2); $\sum f_i = 40$ and $\sum f_i d_i = 0$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{0}{40} = 35$ (1).	2+2+1

	OR: Cumulative frequencies: 11, 13, 23, 36, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 70–80: $l = 70$, $cf = 13$, $f = 10$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{20-13}{10} \times 10 = 77$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 10–20	It belongs to the class 10–20; Modal class = 10–20
2	**(c)** 60–70	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class = 60–70
3	**(c)** 25	$3+5+9+8$; $\sum f_i = 25$
4	**(a)** 45	$17+20+8$; $cf = 45$
5	**(c)** 16.6	$\sum f_i = 25$ and $\sum f_i x_i = 415$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{415}{25} = 16.6$
6	**(c)** 28	$l = 20$, $f_1 = 9$, $f_0 = 5$, $f_2 = 8$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{4}{5} \times 10 = 28$
7	**(a)** 24	Median class 20–30: $l = 20$, $cf = 17$, $f = 20$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{25-17}{20} \times 10 = 24$
8	**(a)** 15	$55 - 40$
9	**(d)** 35	$\frac{30+40}{2}$; Class mark $= 35$
10	**(b)** 40–50	It belongs to the class 40–50; Modal class = 40–50

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Highest frequency is 20, so the modal class is 40–50; $l = 40$, $f_1 = 20$, $f_0 = 17$, $f_2 = 3$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{3}{20} \times 10 = 41.5$ (1).	1+1
12	Cumulative frequencies: 17, 37, 40, 50; $N = 50$, so $\frac{N}{2} = 25$; Median class 30–40: $l = 30$, $cf = 17$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-17}{20} \times 10 = 34$ (1).	1+1
13	Class marks x_i: 35, 45, 55, 65; $\sum f_i = 50$ and $\sum f_i x_i = 2310$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2310}{50} = 46.2$ (1).	1+1
14	Highest frequency is 12, so the modal class is 60–70; $l = 60$, $f_1 = 12$, $f_0 = 5$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{7}{14} \times 10 = 65$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Class marks x_i: 25, 35, 45, 55, 65 (1); $\sum f_i = 50$ and $\sum f_i x_i = 2100$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2100}{50} = 42$ (1).	1+1+1
16	Cumulative frequencies: 19, 24, 26, 48, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 50–60: $l = 50$, $cf = 24$, $f = 2$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{25-24}{2} \times 10 = 55$ (1).	1+1+1
17	Highest frequency is 14, so the modal class is 40–50 (1); $l = 40$, $f_1 = 14$, $f_0 = 8$, $f_2 = 8$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{6}{12} \times 10 = 45$ (1).	1+1+1
	OR: Class marks x_i: 55, 65, 75, 85, 95 (1); $\sum f_i = 50$ and $\sum f_i x_i = 4130$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{4130}{50} = 82.6$ (1).	1+1+1
18	Cumulative frequencies: 15, 24, 28, 45, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 30–40: $l = 30$, $cf = 24$, $f = 4$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-24}{4} \times 10 = 32.5$ (1).	1+1+1
	OR: Highest frequency is 12, so the modal class is 30–40 (1); $l = 30$, $f_1 = 12$, $f_0 = 11$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{1}{8} \times 10 = 31.25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (2); $\sum f_i = 100$ and $\sum f_i d_i = -400$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{-400}{100} = 31$ (1).	2+2+1
	OR: Cumulative frequencies: 8, 11, 31, 34, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 60–70: $l = 60$, $cf = 11$, $f = 20$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{20-11}{20} \times 10 = 64.5$ (1).	2+2+1

20	Take $a = 55$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -2, -1, 0, 1, 2 (2); $\sum f_i = 50$ and $\sum f_i u_i = -8$ (2); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 55 + 10 \times \frac{-8}{50} = 53.4$ (1).	2+2+1
	OR: Highest frequency is 15, so the modal class is 60–70 (2); $l = 60$, $f_1 = 15$, $f_0 = 2$, $f_2 = 2$, $h = 10$ (2); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{13}{26} \times 10 = 65$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 60–70	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class = 60–70
2	**(c)** 50	$11+16+13+2+8$; $\sum f_i = 50$
3	**(b)** 8	$4+4$; $cf = 8$
4	**(b)** 46.2	$\sum f_i = 50$ and $\sum f_i x_i = 2310$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2310}{50} = 46.2$
5	**(d)** 26.25	$l = 20$, $f_1 = 16$, $f_0 = 11$, $f_2 = 13$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{5}{8} \times 10 = 26.25$
6	**(d)** 32.5	Median class 30–40: $l = 30$, $cf = 8$, $f = 8$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{10-8}{8} \times 10 = 32.5$
7	**(d)** mode	Definition of mode
8	**(a)** 45	$\frac{40+50}{2}$; Class mark = 45
9	**(b)** 20–30	It belongs to the class 20–30; Modal class = 20–30
10	**(d)** 50–60	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class = 50–60

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cumulative frequencies: 11, 31, 35, 50; $N = 50$, so $\frac{N}{2} = 25$; Median class 30–40: $l = 30$, $cf = 11$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-11}{20} \times 10 = 37$ (1).	1+1
12	Class marks x_i: 15, 25, 35, 45; $\sum f_i = 50$ and $\sum f_i x_i = 1610$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1610}{50} = 32.2$ (1).	1+1
13	Highest frequency is 9, so the modal class is 60–70; $l = 60$, $f_1 = 9$, $f_0 = 8$, $f_2 = 2$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{1}{8} \times 10 = 61.25$ (1).	1+1
14	Cumulative frequencies: 6, 13, 33, 40, 50; $N = 50$, so $\frac{N}{2} = 25$; Median class 50–60: $l = 50$, $cf = 13$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{25-13}{20} \times 10 = 56$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Class marks x_i: 45, 55, 65, 75, 85 (1); $\sum f_i = 50$ and $\sum f_i x_i = 3160$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{3160}{50} = 63.2$ (1).	1+1+1
16	Cumulative frequencies: 4, 21, 26, 34, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 20–30: $l = 20$, $cf = 21$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{25-21}{5} \times 10 = 28$ (1).	1+1+1
17	Highest frequency is 17, so the modal class is 60–70 (1); $l = 60$, $f_1 = 17$, $f_0 = 4$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{13}{25} \times 10 = 65.2$ (1).	1+1+1
	OR: Class marks x_i: 35, 45, 55, 65, 75 (1); $\sum f_i = 50$ and $\sum f_i x_i = 2660$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2660}{50} = 53.2$ (1).	1+1+1
18	Cumulative frequencies: 5, 7, 17, 19, 25; $N = 25$, so $\frac{N}{2} = 12.5$ (1); Median class 60–70: $l = 60$, $cf = 7$, $f = 10$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{12.5-7}{10} \times 10 = 65.5$ (1).	1+1+1
	OR: Highest frequency is 20, so the modal class is 50–60 (1); $l = 50$, $f_1 = 20$, $f_0 = 8$, $f_2 = 7$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{12}{25} \times 10 = 54.8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take $a = 75$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -2, -1, 0, 1, 2 (2); $\sum f_i = 50$ and $\sum f_i u_i = 2$ (2); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 75 + 10 \times \frac{2}{50} = 75.4$ (1).	2+2+1
	OR: Highest frequency is 20, so the modal class is 30–40 (2); $l = 30$, $f_1 = 20$, $f_0 = 2$, $f_2 = 8$, $h = 10$ (2); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{18}{30} \times 10 = 36$ (1).	2+2+1

20	Take $a = 45$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (2); $\sum f_i = 100$ and $\sum f_i u_i = -19$ (2); Mean $= a + h\,\frac{\sum f_i u_i}{\sum f_i} = 45 + 10 \times \frac{-19}{100} = 43.1$ (1).	2+2+1
	OR: Highest frequency is 16, so the modal class is 80–90 (2); $l = 80$, $f_1 = 16$, $f_0 = 10$, $f_2 = 7$, $h = 10$ (2); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{6}{15} \times 10 = 84$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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